

Olist E-Commerce SQL Project (Advanced Business Case)

Business Context

Olist is a multi-vendor e-commerce platform operating in Brazil, connecting customers with sellers across regions. The platform manages order lifecycle, payments, logistics, and customer reviews. This dataset simulates real-world business operations where multiple stakeholders interact in a transactional ecosystem.

Problem Statement

Despite having large volumes of transactional data, the business lacks structured insights into revenue drivers, customer satisfaction, and operational efficiency. There is a need to transform raw data into actionable insights that can guide decision-making.

Key Business Questions

1. What are the overall revenue trends and growth patterns? 2. Which products and sellers contribute the most to revenue? 3. Are delivery delays impacting customer satisfaction? 4. Which payment methods drive the highest value transactions? 5. How does customer location influence sales and logistics?

Key Performance Indicators (KPIs)

• Total Revenue • Average Order Value (AOV) • Order Volume • Delivery Time (Actual vs Estimated) • Customer Satisfaction Score (Review Ratings) • Top Performing Sellers & Products

Dataset Overview

The dataset consists of 9 relational tables: customers, orders, order_items, products, sellers, payments, reviews, geolocation, and category translation. These tables collectively represent the end-to-end e-commerce workflow.

Project Objectives

• Perform data cleaning and validation • Build relational data model using SQL joins • Generate business insights using SQL queries • Create reusable views and stored procedures • Prepare data for visualization in Power BI

Analytical Approach

Step 1: Data Quality Checks (NULLs, duplicates, integrity) Step 2: Data Cleaning and transformation Step 3: Exploratory Data Analysis using SQL Step 4: Business insight generation Step 5: Dashboard integration with Power BI

Expected Outcome

A complete end-to-end SQL project demonstrating strong data handling, business understanding, and analytical thinking. The project will provide actionable insights that simulate real-world business decision-making.